Ethical Hacking

Rubric 5

Basic Exploit

Basic Exploitation

The project must demonstrate basic exploitation skills in a controlled lab environment using Metasploit Framework with proper target verification and scope definition. Exploitation process must be documented step-by-step including tool configurations, execution steps, and results. All activities must follow safety guidelines and ethical considerations with proper documentation. Evidence of proper lab containment and clean-up procedures must be included. A comprehensive report detailing the exploitation process must be provided.

"Obtain the pfSense installation image"
Download the ISO from:

https://www.pfsense.org/download/

Pick **AMD64** architecture, **DVD Image (ISO) Installer**. You'll get a file like `pfSense-CE-2.7.x-RELEASE-amd64.iso`.

"Create the firewall virtual machine in VirtualBox"
VirtualBox Manager → **New**:

- Name: `pfSense-Firewall`
- Type: **BSD**
- Version: **FreeBSD (64-bit)**

"Allocate virtual hardware"

- RAM: 1–2 GB
- CPU: 1 core
- Disk: new virtual disk, 8–20 GB, dynamically allocated

"Prepare the VM for installation"
Settings → Storage → click the empty optical drive → Choose a disk file → select the pfSense ISO you downloaded. This is what lets the VM boot into the installer.

That's the full scope of Day 1 — don't install it yet (that's Day 2) and don't touch network adapters yet (that's Day 3, when you map WAN/LAN/DMZ).

Screenshots to capture for this task:

- The pfSense download page with the version visible
- VirtualBox's New VM wizard showing name/type/version
- The hardware allocation screen (RAM, CPU, disk)
- Storage settings showing the ISO mounted on the optical drive

Go ahead and work through it — let me know once the ISO is attached and ready, or if anything in VirtualBox doesn't match what I described (older VirtualBox versions sometimes phrase the wizard steps slightly differently).